QUALIFICATION REQUIREMENTS RE[illegible]VED

MIL-M-38510/481A(USAF)
18 June 1984
SUPERSEDING
MIL-M-38510/481(USAF)
28 February 1982

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, N-CHANNEL, SILICON GATE
MONOLITHIC DUAL SERIAL INPUT/OUTPUT CONTROLLER

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION.

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic -channel, silicon gate dual serial input/output controller. One product assurance lass and a choice of case outlines and lead finishes are provided, and are reflected) the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, with ie exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device types. The device types shall be as follows:

Device type	Clock frequency	Circuit
01	4.0 MHz	Dual serial input/ output controller
02	2.5 MHz	

1.2.2 Device class. The device class shall be the product assurance level as fined in MIL-M-38510.

1 Case outline. The case outline shall be designated as follows:

Letter	Case outline, see MIL-M-38510, appendix C
Q	D-5 (40-lead, 9/16" x 2"), dual-in-line package

1.3 Absolute maximum ratings.

V_{CC} supply voltage range (V_{CC} - GND)	-0.3 V dc to +7 V dc
Voltage on any pin (referenced to ground)	-0.3 V dc to +7 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation at -55°C	1.0 W
Maximum power dissipation at +125°C	0.5 W
Lead temperature (soldering - 5 seconds)	+270°C
Maximum junction temperature (T_J)	+170°C
Thermal resistance, junction-to-case (θ_{JC})	
Case Q	(See appendix C of MIL-M-38510)

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc (minimum) to 5.5 V dc (maximum)
Minimum high-level input voltage (V_{IH})	
Logic inputs	2.0 V dc
Clock input	V_{CC} -0.6 V dc

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, RADC (RBE-2), Griffiss AFB, NY 13441, by using the [illegible]-addressed Standardization Document Improvement Proposal (DD Form 1426) a[illegible]aring at the end of this document or by letter.

FSC 5962

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Maximum low-level input voltage (V_{IL})	
Logic inputs	0.8 V dc
Clock input	0.45 V dc
Frequency of operation (01 circuit)	0.5 to 4.0 MHz
(02 circuit)	0.5 to 2.5 MHz
Case operating temperature range	-55°C to +125°C
Clock rise time (device types 01 and 02)	30 ns maximum
Clock fall time (device types 01 and 02)	30 ns maximum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification For.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. In the event of conflict between MIL-M-38510 and this detail specification, this detail specification shall govern.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections and pin assignments. Terminal connections shall be as specified on figure 1.

3.2.2 Functional block diagram. The functional block diagram shall be as specified on figure 2.

3.2.3 Functional description, terms, and symbols. Functional description, terms, and symbols shall be as specified in 6.4.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and on figure 6, and apply over the full recommended case operating temperature range, unless otherwise specified.

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TABLE I. Electrical performance characteristics.

Parameter	Symbol	Conditions (T_C = -55°C to +125°C; V_{CC} = 5.0 V ±10% unless otherwise specified)	Figure number	Reference number	01		02		Limits		Unit
					Min	Max	Min	Max	Min	Max	
Clock input low voltage	V_{IL1}								-0.3	0.45	V
Clock input high voltage	V_{IH1}								V_{CC} - 0.6 V	5.5	V
Input low voltage	V_{IL2}								-0.3	0.8	V
Input high voltage	V_{IH2}								2.0	5.5	V
Output low voltage	V_{OL}	I_{OL} = 2.0 mA								0.4	V
Output high voltage	V_{OH}	I_{OH} = -250 µA							2.4		V
Power supply current	I_{CC}	Outputs open (V_{CC} = 5.5 V, T_C = -55°C)								182	mA
Output leakage current low, open drain outputs	I_{LOL}	V_{IN} = 0.4 V							-10	+10	µA
Output leakage current high, open drain outputs	I_{LOH}	V_{IN} = 2.4 V							-10	+10	µA
$\overline{SYNCA}$ pin leakage current low	I_{IL1}	V_{IN} = 0.4 V							-40	+10	µA
$\overline{SYNCA}$ pin leakage current high	I_{IH1}	V_{IN} = 2.4 V							-40	+10	µA
Input low current (input and bidirectional	I_{IL2}	V_{IN} = 0.4 V							-10	+10	µA
Input high current (input and bidirectional	I_{IH2}	V_{IN} = 2.4 V							-10	+10	µA

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TABLE I. Electrical performance characteristics - Continued.

Parameter		Symbol	Conditions (T_C = -55°C to +125°C; V_{CC} = 5.0 V ±10% unless otherwise specified)	Figure number 5/	Reference number 4/	01		02		Limits		Unit
						Min	Max	Min	Max	Min	Max	
Maximum frequency	01	f_{MAX}	4.5 V ≤ V_{CC} ≤ 5.5 V							4.0		MHz
	02		C_L = 50 pF ±10%							2.5		MHz
Input clock capacitance		C_{I1}	T_C = +25°C								40	pF
Input capacitance, all other inputs		C_{I2}	T_C = +25°C								5	pF
Output capacitance, all outputs and data bus		C_O	T_C = +25°C								10	pF
Clock cycle time 2/, 3/		T_{CYC1}	C_L = 50 pF ±10%; -55°C ≤ T_C ≤ +125°C	6	1	250	2000	400	2000			ns
$\overline{TxC}$ cycle time		T_{CYC2}			24	400	∞	400	∞			
$\overline{RxC}$ cycle time		T_{CYC3}			30	400	∞	400	∞			
Clock fall time		t_{fC}			3		30		30			
Clock rise time		t_{rC}			4		30		30			
Clock width high		t_{PWH1}	6/		2	105		170				
Clock width low		t_{PWL1}	7/		5	105		170				
Pulse width high ($\overline{CTS}$, $\overline{DCD}$, $\overline{SYNC}$)		t_{PWH2}	Use figure 4 load		22	200		200				
Pulse width low ($\overline{CTS}$, $\overline{DCD}$, $\overline{SYNC}$)		t_{PWL2}			23	200		200				
$\overline{TxC}$ width high		t_{PWH3}			26	180	∞	180	∞			

See footnotes at end of table.

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Parameter	Symbol	Conditions (T_C = -55°C to +125°C; V_{CC} = 5.0 V ±10% unless otherwise specified)	Fig. number 5/	Reference number 4/	01		02		Limits		Unit
					Min	Max	Min	Max	Min	Max	
$\overline{TxC}$ width low	t_{PWL3}	C_L = 50 pF ±10% -55°C ≤ T_C ≤ +125°C	6	25	180	∞	180	∞			ns
$\overline{RxC}$ width high	t_{PWH4}			32	180	∞	180	∞			
$\overline{RxC}$ width low	t_{PWL4}			31	180	∞	180	∞			
$\overline{CE}$, C/$\overline{D}$, B/$\overline{A}$ to clock ↑ setup	t_{SHL1} t_{SLH1}			6	145		160				
$\overline{IORQ}$, $\overline{RD}$ to clock ↑ setup	t_{SHL2} t_{SLH2}			7	115		240				
Data in to clock ↑ setup (Write or $\overline{M1}$ cycle)	t_{SHL3} t_{SLH3}			9	50		50				
RxD to $\overline{RxC}$ ↑ setup (x1 mode)	t_{SHL4} t_{SLH4}			33	0		0				
$\overline{M1}$ to clock ↑ setup	t_{SHL5} t_{SLH5}			12	90		210				
IEI to $\overline{IORQ}$ ↓ setup (INTA cycle)	t_{SHL6} t_{SLH6}			13	140		200				
$\overline{SYNC}$ ↓ to $\overline{RxC}$ ↑ setup (External SYNC modes)	t_{SHL7} t_{SLH7}			38	100		-100				
$\overline{RxC}$ ↑ to $\overline{RxD}$ hold (x1 mode)	t_{HHL1} t_{HLH1}			34	140		140				
Any unspecified hold when setup specified	t_{HHL2} t_{HLH2}			21	0		0				

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Parameter	Symbol	Conditions (T_C = -55°C to +125°C; V_{CC} = 5.0 V ±10% unless otherwise specified)		Figure number 5/	Reference number 4/	01		02		Limits		Unit
						Min	Max	Min	Max	Min	Max	
Clock ↑ to data out delay	t_{PHL1} t_{PLH1}	C_L = 50 pF ±10% -55°C ≤ T_C ≤ +125°C	Use figure 4 load	6	8		220		240			ns
$\overline{RD}$ ↑ to data out float delay	t_{PHZ1} t_{PLZ1}				10		110		230			
$\overline{IORQ}$ ↓ to data out delay (INTA cycle)	t_{PHL2} t_{PLH2}				11		160		340			
$\overline{TxC}$ ↓ to TxD delay (x1 mode)	t_{PHL3} t_{PLH3}				27		300		400			
$\overline{M1}$ ↓ to IEO ↓ delay (Interrupt before M1)	t_{PHL4}				14	190		300				
IEI ↑ to IEO ↑ (After ED decode)	t_{PLH4}				15		100		150			
IEI ↓ to IEO ↓ delay	t_{PHL5}				16		100		150			
Clock ↑ to $\overline{INT}$ ↓ delay	t_{PHL6}		Use figure 5 load		17		200		200			
$\overline{IORQ}$ ↓ or $\overline{CE}$ ↓ to $\overline{W/RDY}$ delay (Wait mode)	t_{PHL7}		Use figure 4 load		18		210		300			
Clock ↑ to $\overline{W/RDY}$ ↓ delay (Ready mode)	t_{PHL8}		Use figure 5 load		19		120		120			
Clock ↓ to $\overline{W/RDY}$ float delay (Wait mode)	t_{PLZ2}		Use figure 5 load		20		130		150			

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Parameter	Symbol	Conditions (T_C = -55°C to +125°C; V_{CC} = 5.0 V ±10% unless otherwise specified)		Figure number 5/	Reference number 4/	01		02		Limits		Unit
						Min	Max	Min	Max	Min	Max	
$\overline{TxC}$ ↓ to $\overline{W/RDY}$ ↓ delay (Ready mode)	t_{PHL9}	C_L = 50 pF ±10% -55°C ≤ T_C ≤ +125°C	Use figure 5 load	6	28	5	9	5	9			1/
$\overline{TxC}$ ↓ to $\overline{INT}$ ↓ delay	t_{PHL10}				29	5	9	5	9			
$\overline{RxC}$ ↑ to $\overline{W/RDY}$ ↓ delay (Ready mode)	t_{PHL11}				35	10	13	10	13			
$\overline{RxC}$ ↑ to $\overline{INT}$ ↓ delay	t_{PHL12}				36	10	13	10	13			
$\overline{RxC}$ ↑ to $\overline{SYNC}$ ↓ delay (Output modes)	t_{PHL13}		Use figure 4 load		37	4	7	4	7			

NOTES:

1/ Clock periods (system clock).

2/ In all modes, the system clock rate must be at least five times the maximum data rate.

3/ RESET must be active a minimum of one complete clock cycle.

4/ The reference number refers to the position where the parameters being tested appears on figure 6.

5/ Figure 6 represents the electrical performance characteristics associated with the AC limits shown in table I.

6/ $t_{PWH1} = t_{CYC} - t_{PWL1} - t_{RC} - t_{fC}$

7/ $t_{PWL1} = t_{CYC} - t_{PWH1} - t_{RC} - t_{fC}$

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IV.	Input/output conditions for subgroups 9, 10, and 11.	30
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VI.	Test vectors	(see 6.1)

3.5 Electrical test requirements. Electrical test requirements shall be as specified in tables III and IV, and figure 6, for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, and quality conformance by device class are specified in table II.

3.6 Register definitions. Register definitions shall be as specified in table V.

3.7 Test vectors. Test vectors shall be as specified in table VI (see 6.1).

3.8 Marking. Marking shall be in accordance with MIL-M-38510. The "JAN" or "J" certification mark shall not be used.

3.9 Microcircuit group assignment. The devices covered by this specification shall be in a microcircuit group number 107 (see MIL-M-38510, appendix E).

3.10 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have manufacturer certification in accordance with MIL-M-38510 for at least one line; not necessarily the line producing the device types described herein.

3.11 Certification. Certification in accordance with MIL-M-38510 is not required for these devices.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirement	Subgroups (see table III)
	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1, 7
Final electrical test parameters (method 5004)	*1, 2, 3, *7, 8, 9**, 10, 11
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9**, 10, 11
Group B zap test	See 4.5.2
Group C end-point electrical parameters (method 5005)	1, 2, 3, 7, 8
Additional electrical subgroups for group C periodic inspection	None

*PDA applies to subgroups 1 and 7 (see 4.2c).
**Subgroup 9 is run concurrently with subgroup 7 with a combined LTPD of 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

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a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.
(2) T_A = +125°C.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

d. Constant acceleration in accordance with test condition B of method 2001 of MIL-STD-883.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in ccordance with MIL-M-38510. Inspections to be performed shall be those specified n method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see .4.1 through 4.4.3). Generic test data (see 6.7) may be used to satisfy the or ements for groups C and D inspections. Quality conformance inspection shall e pleted on the specific devices covered by this specification before they are hipped.

4.4.1 Group A inspection. Group A inspection shall consist of the test subgroups nd LTPD values shown in table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II, herein.

b. Subgroup 4 (C_{I1}, C_O, C_{I2} measurement) shall be performed only after process or design changes which may affect device capacitance.

c. Subgroup 12 shall be added to group A inspection requirements for class B devices using an LTPD of 10 and consist of the procedures, test conditions, and limits specified in table III. This subgroup shall be used for initial verification only and variables data shall be submitted.

4.4.2 Group B inspection. Group B inspection shall consist of the test subgroups nd LTPD values shown in table II of method 5005 of MIL-STD-883, and as follows:

A special subgroup shall be added to the group B inspection requirements, using an LTPD of 15 for class B. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.2).

4.4.3 Groups C and D inspections. Groups C and D inspections shall consist of he test subgroups and LTPD values shown in tables III and IV of method 5005 of STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II, herein.

b. Steady-state life test (method 1005 of MIL-STD-883).

(1) Test condition D or E using the circuit shown on figure 3.
(2) T_A = +125°C, minimum.

c. Constant acceleration in accordance with test condition B of method 2001 of MIL-STD-883.

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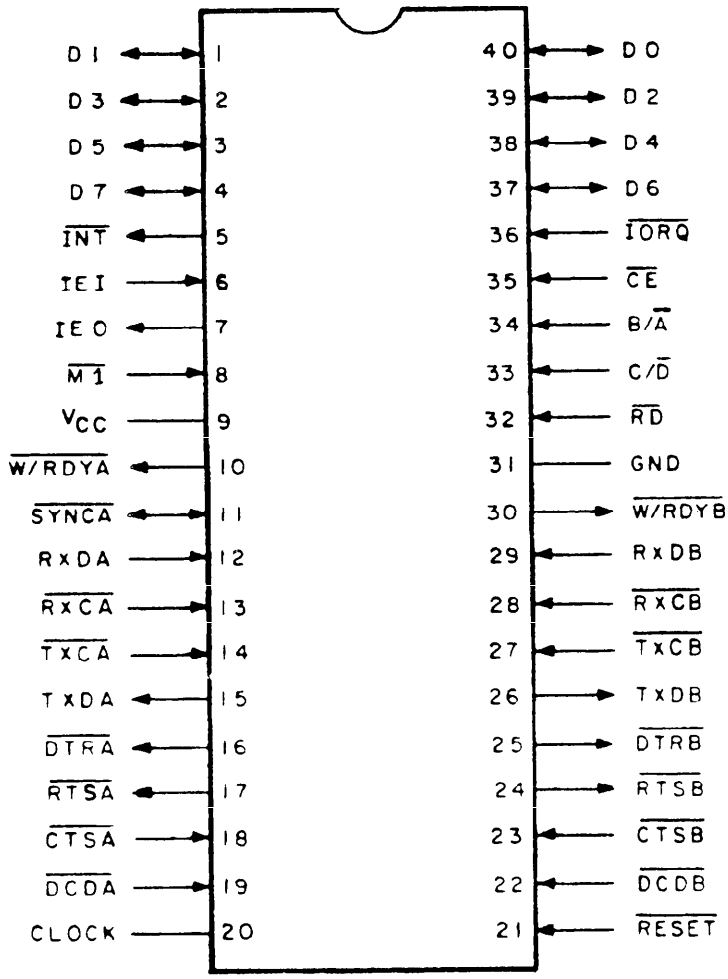


FIGURE 1. Terminal connections.

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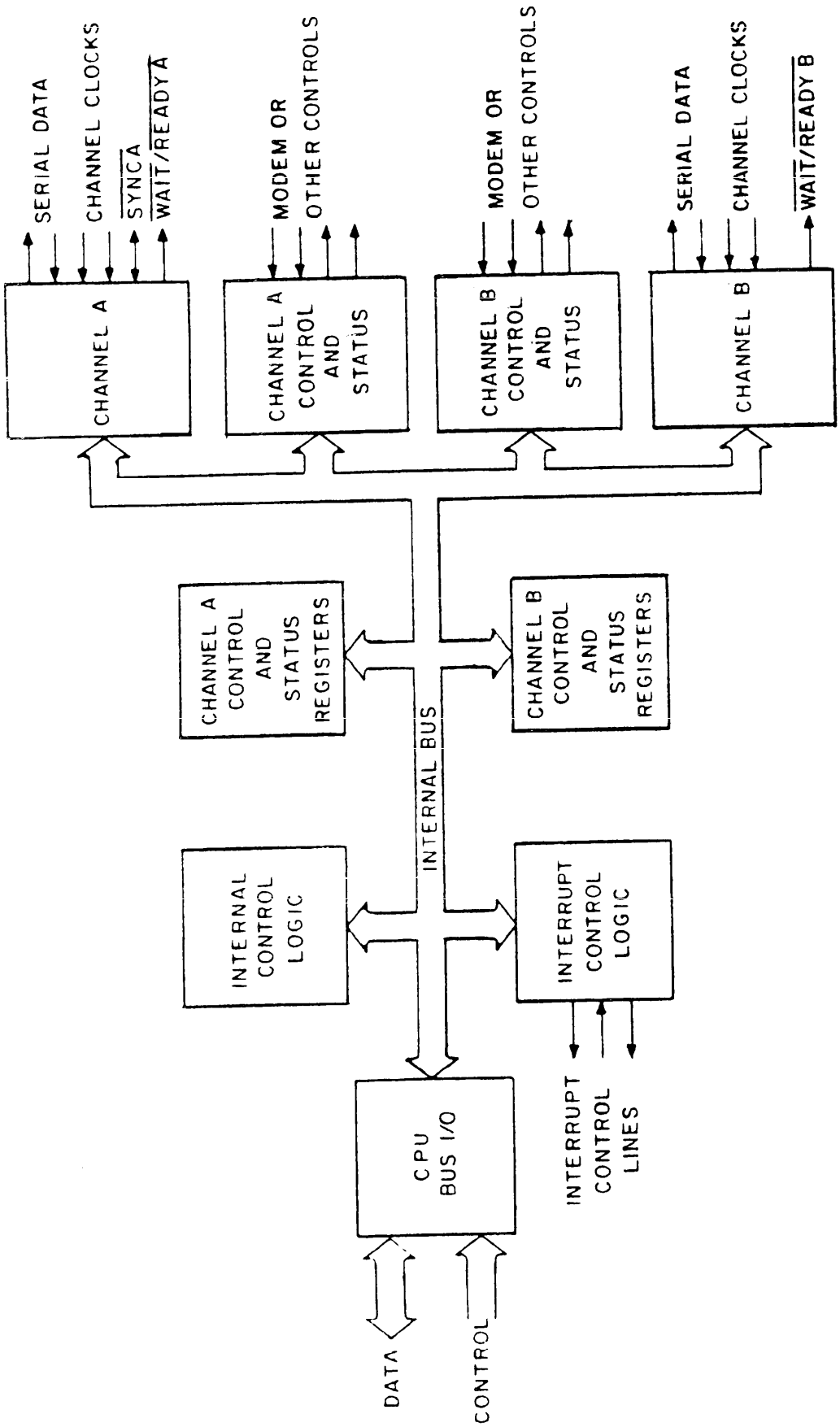
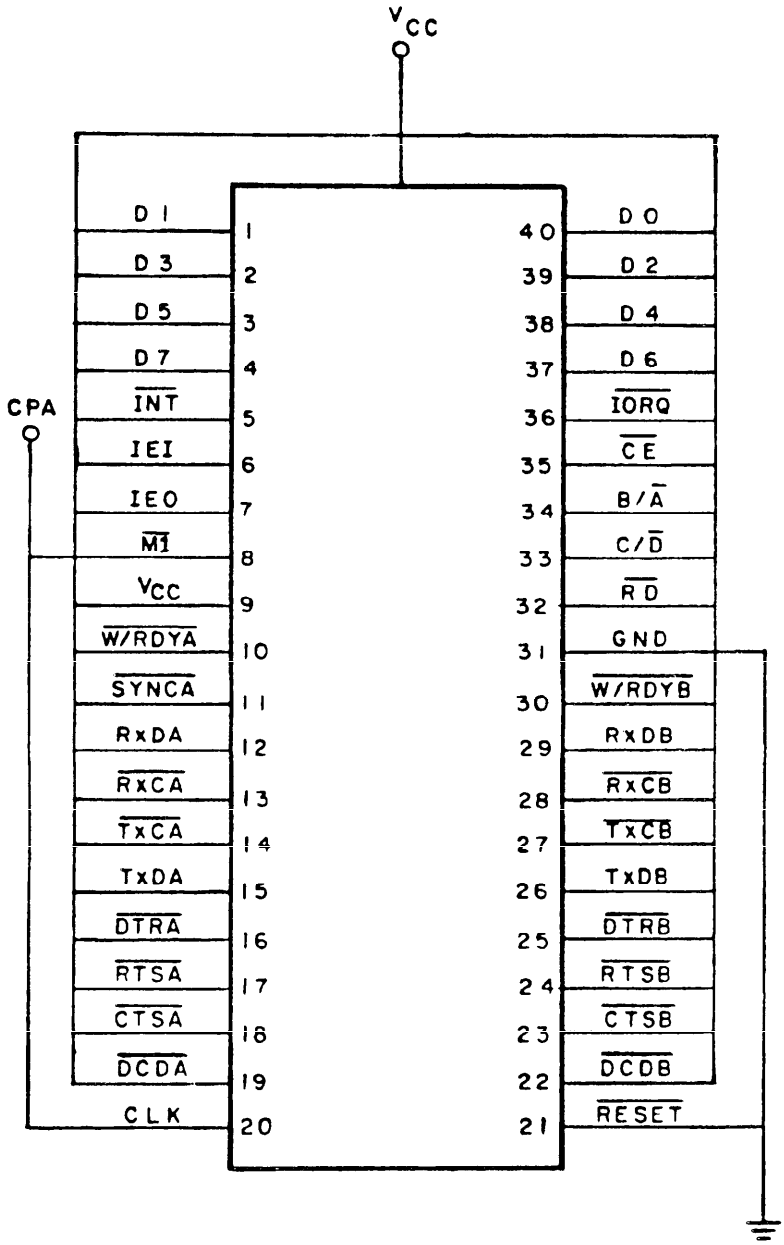


FIGURE 2. Functional block diagram.

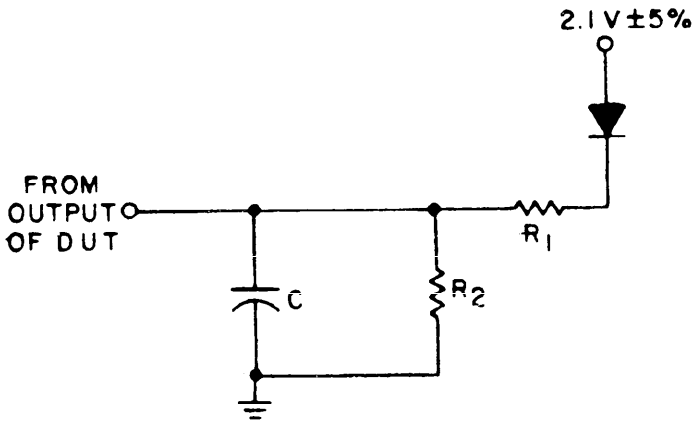
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NOTES:
1. CPA = 100 kHz ±10% square wave.
   50% duty cycle
2. V_{CC} shall be no less than 4.5 V at device terminal.
   V_{CC} shall be no more than 5.5 V at device terminal.
3. V_{ILC} shall be no more than 0.5 V at CPA.
   V_{IHC} shall be no less than 3.9 V at CPA.

FIGURE 3. Burn-in and life test circuit.

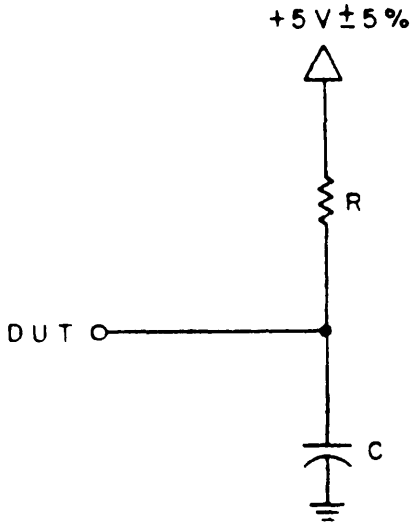
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NOTES:
1. C = 50 pF ±10% includes tester, wiring and stray capacitance.
2. R_1 = 510Ω ±5%, R_2 = 9.53 kΩ ±1%.
3. The diode is 1N3064 or equivalent.

FIGURE 4. Output load circuit all outputs not open drain, for ac functional testing (all device types).

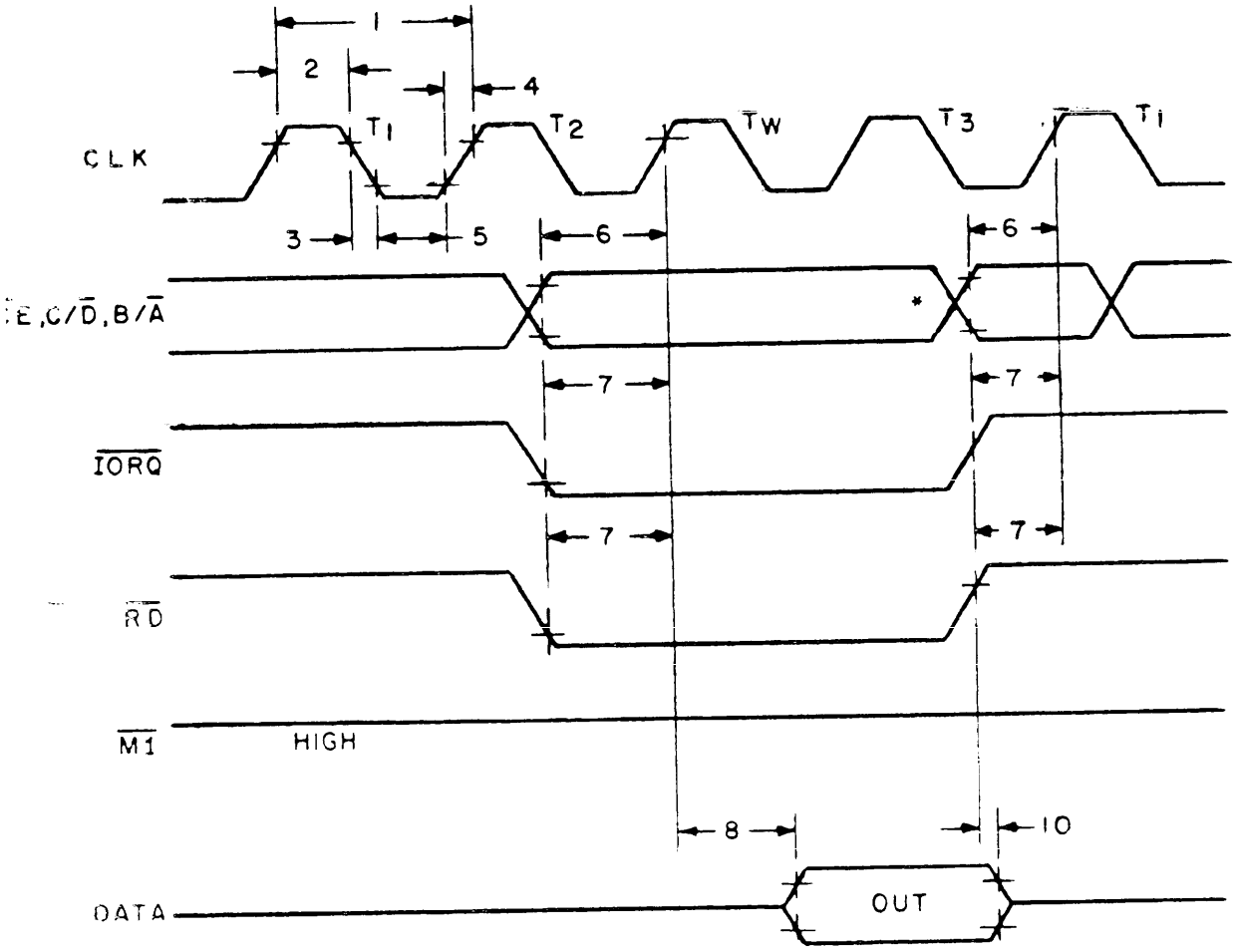
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NOTES:
1. C = 50 pF ±10% including tester, wiring and stray capacitance.
2. R = 2.2 kΩ ±5%.
3. Open drain outputs are INT, $\overline{W/RDYA}$, $\overline{W/RDYB}$.

FIGURE 5. Output load circuit for open drain outputs for ac and dc functional testing, both device types.

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* Optional condition occurs only when any combination of these inputs must change state to reprogram during read/write operation.

FIGURE 6. Read cycle.

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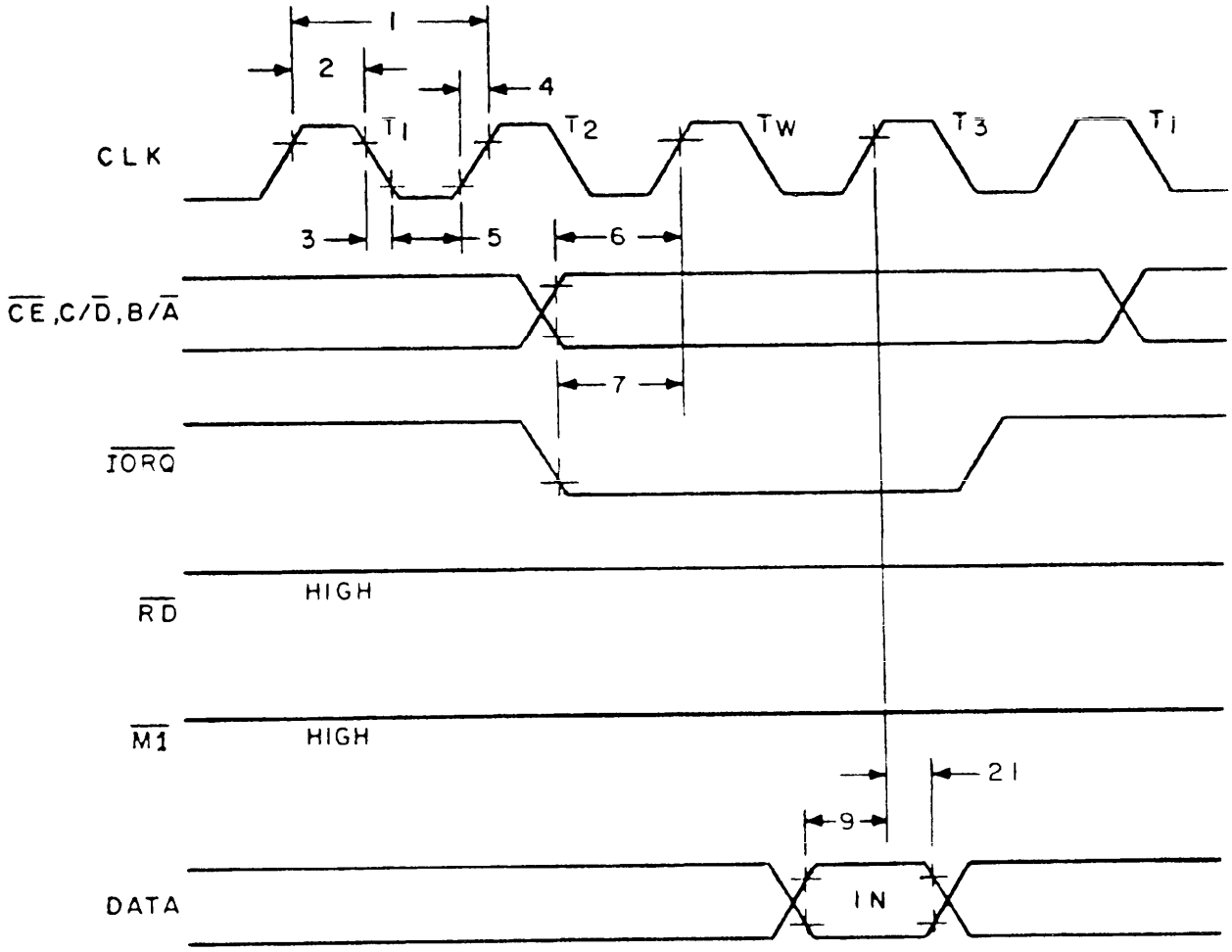


FIGURE 6. Write cycle - Continued.

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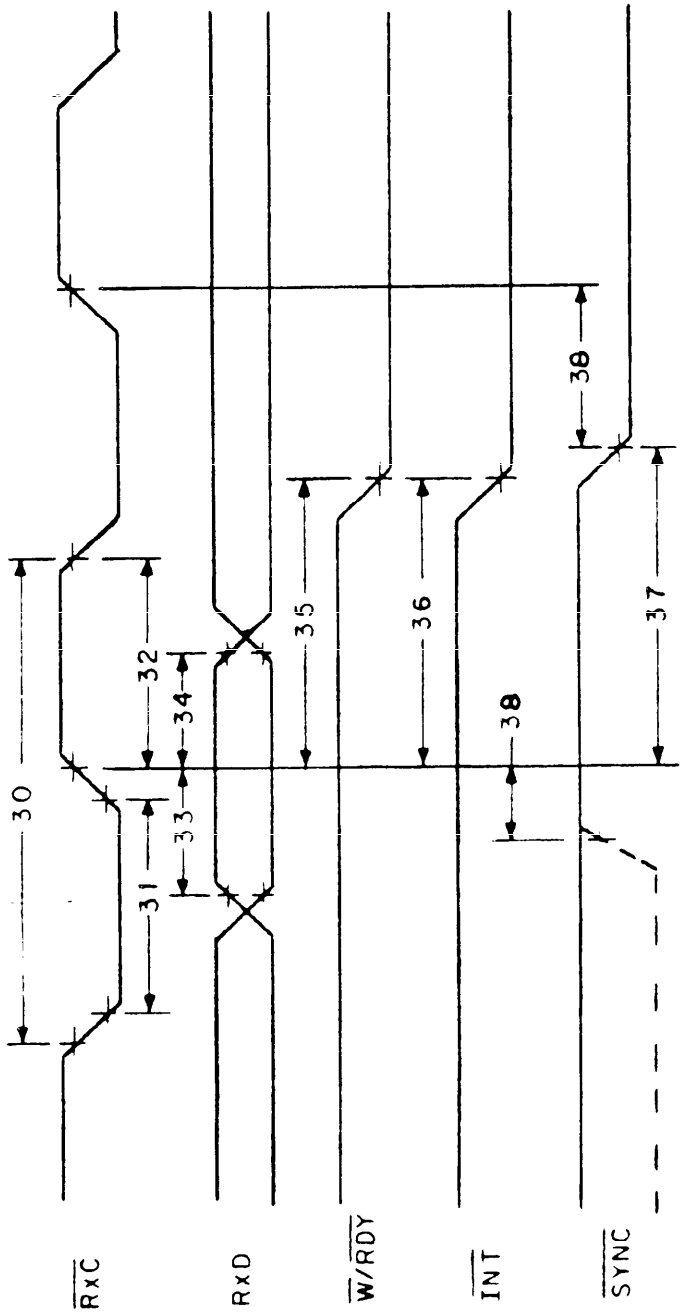


FIGURE 6. Receiver serial data - Continued.

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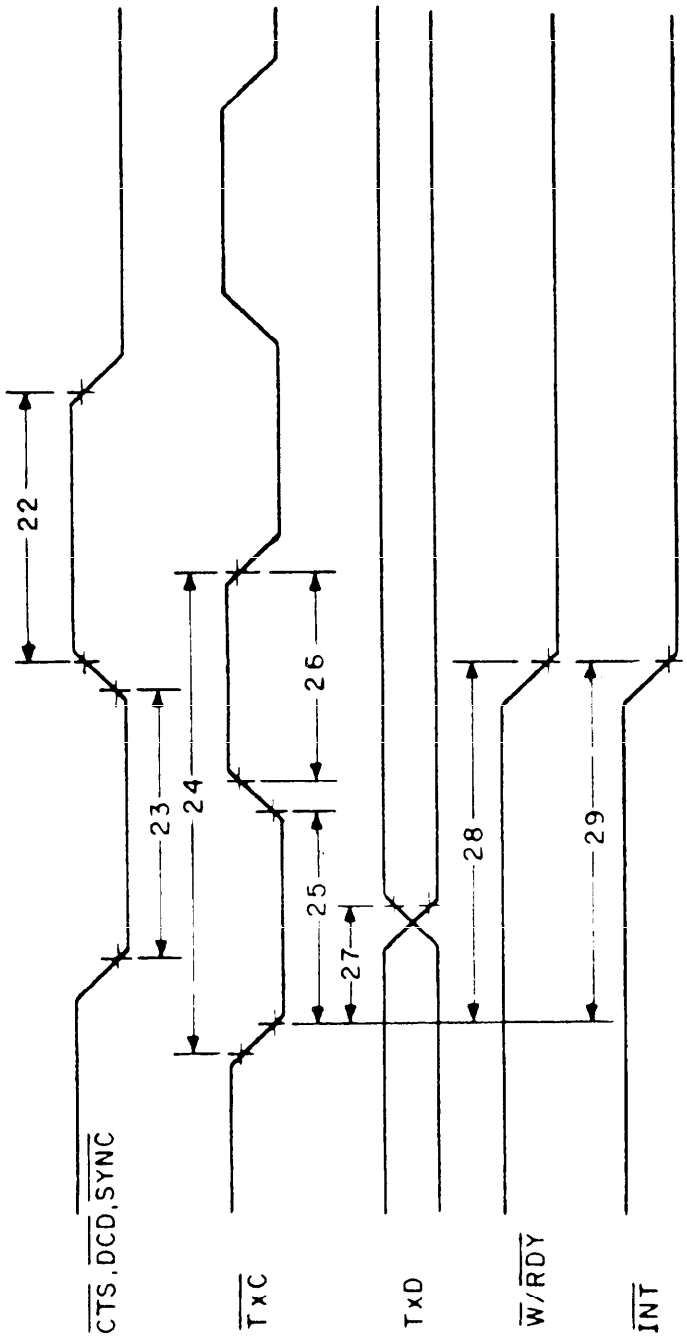


FIGURE 6. Transmitter serial data and modern control - Continued.

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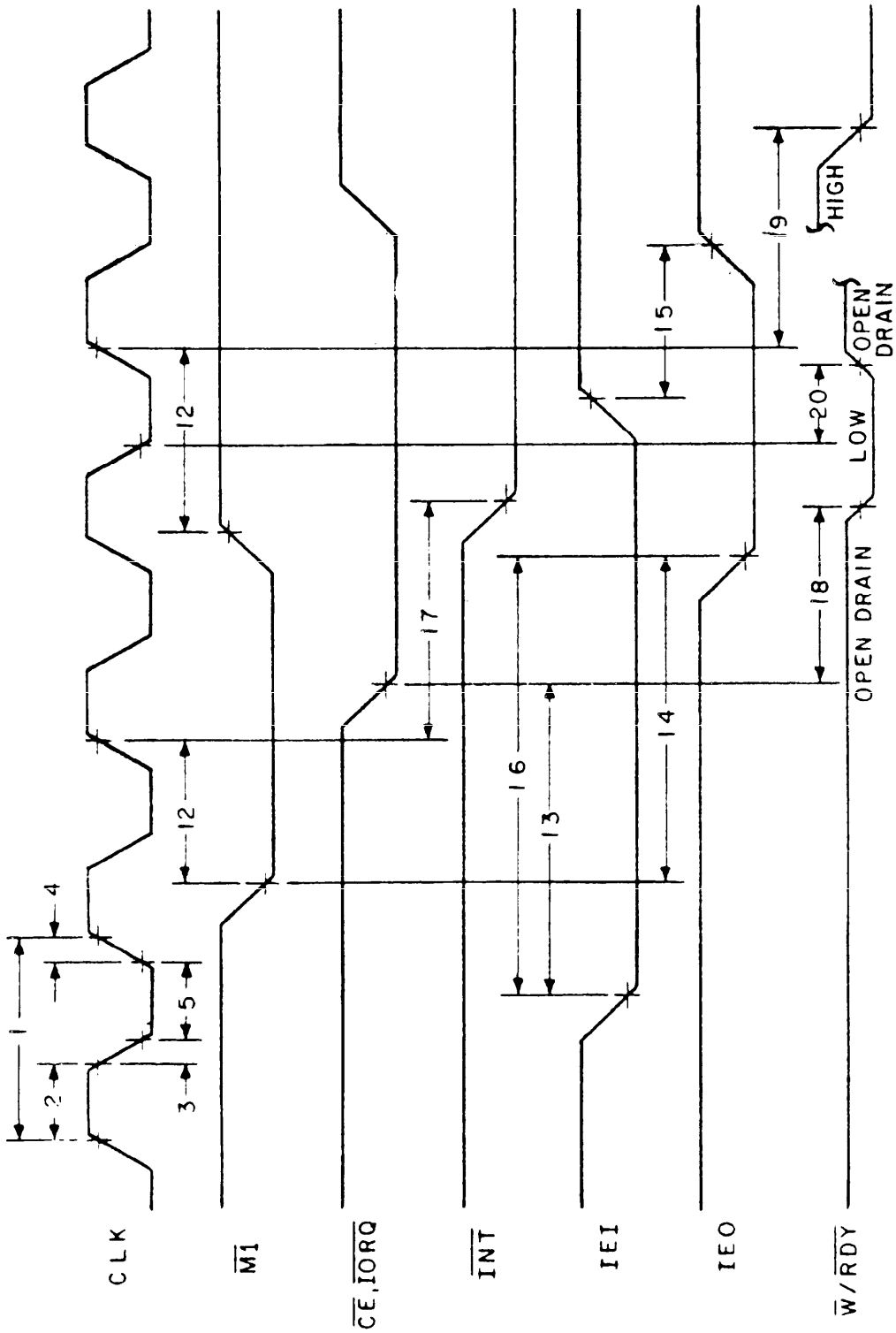


FIGURE 6. Interrupt request - Continued.

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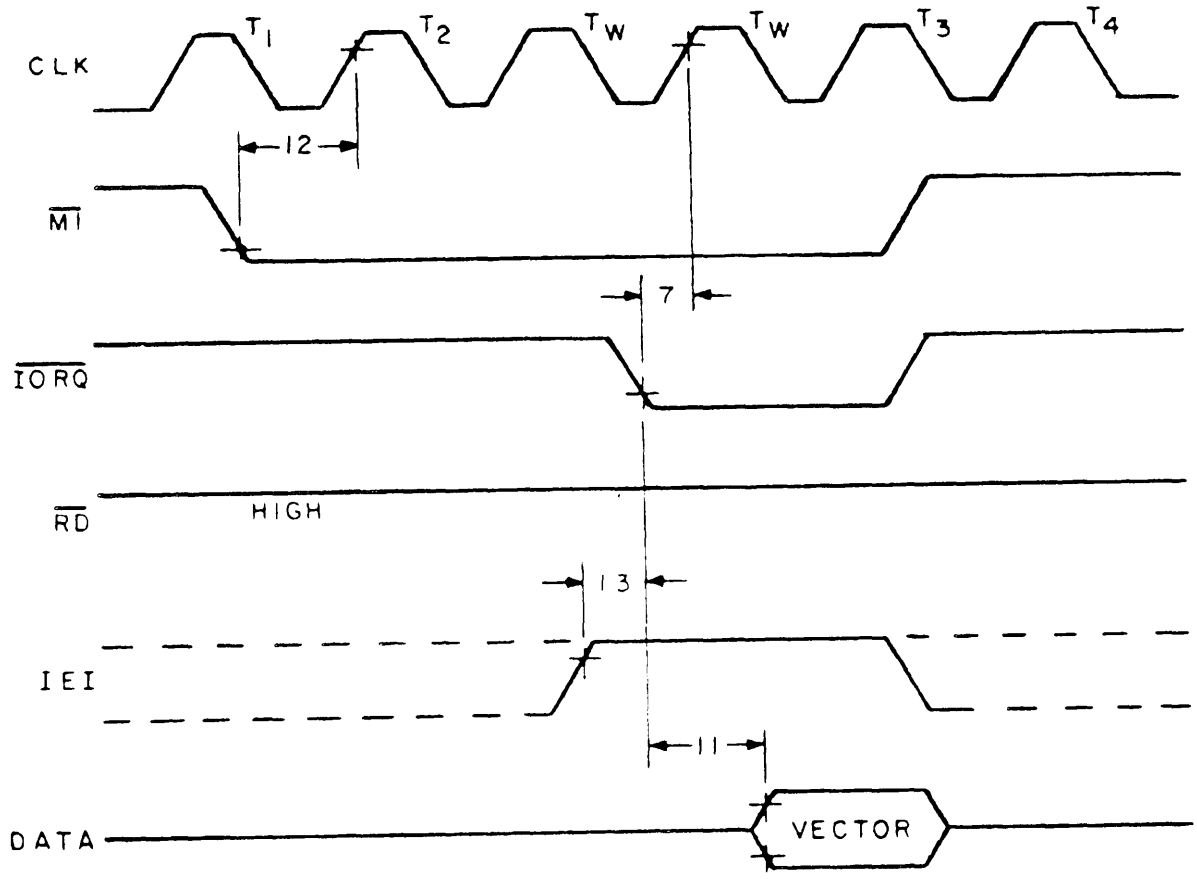


FIGURE 6. Interrupt acknowledge cycle - Continued.

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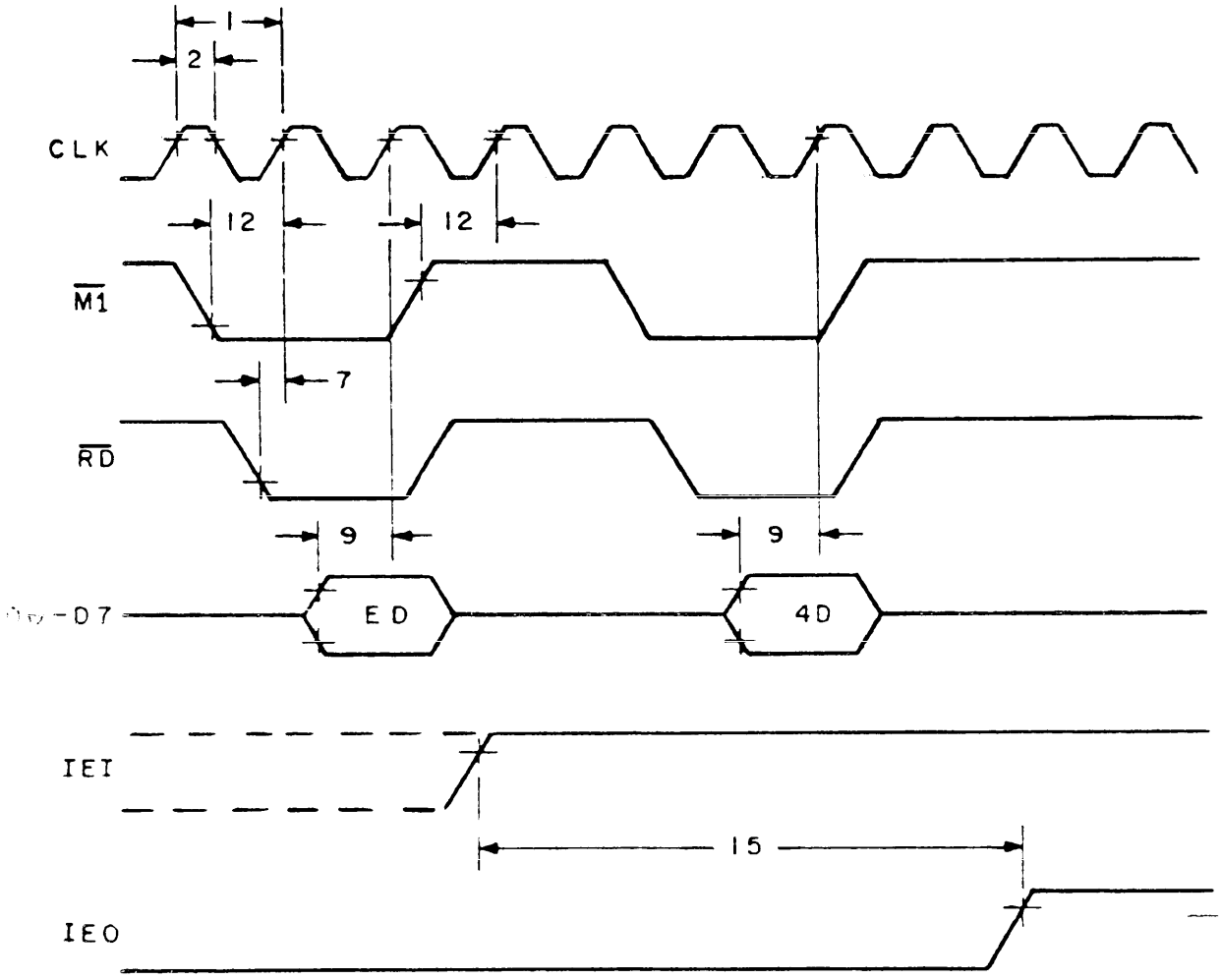


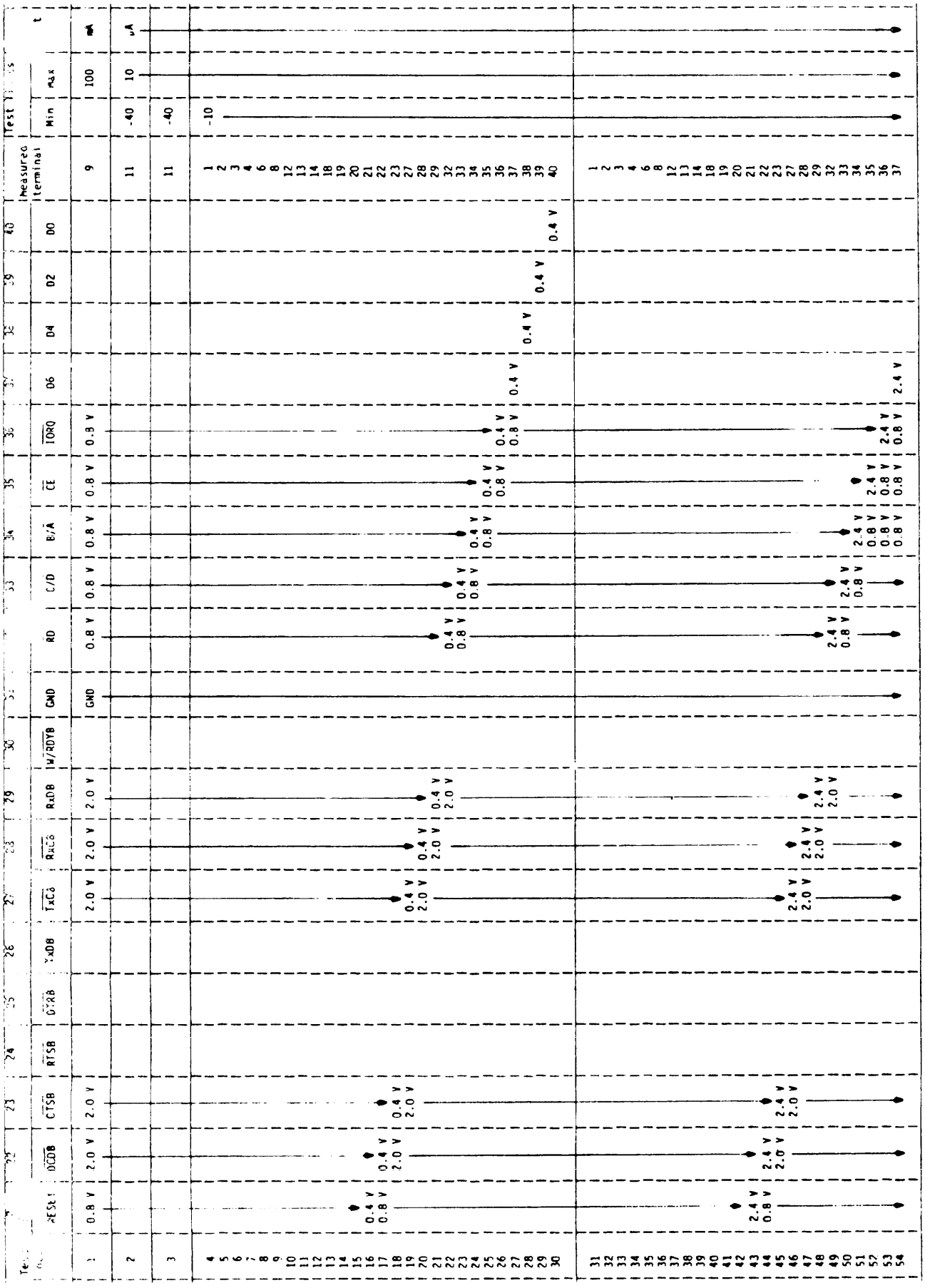
FIGURE 6. Return from interrupt cycle - Continued.

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TABLE III. Group A inspection for device types 01 and 02.
Terminal conditions (pins not designated are open).

Subgroup	Symbol	MIL-STD-883 method	Test no.	1 D1	2 D3	3 D5	4 D7	5 $\overline{INT}$	6 IEI	7 IEO	8 $\overline{MI}$	9 V_{CC}	10 IM/RDYA	11 $\overline{STMCA}$	12 RxDA	13 RxCA	14 $\overline{TxCA}$	15 TxDA	16 $\overline{DTRA}$	17 $\overline{RTSA}$	18 $\overline{CTSA}$	19 $\overline{DCDA}$	20 CLK
1 T_C = +25°C	I_{CC}	3005	1						0.8 V		0.8 V	5.5 V		2.0 V	2.0 V	2.0 V	2.0 V				2.0 V	2.0 V	4.9 V
	I_{IL1}		2											0.4 V									
	I_{IH1}		3											2.4 V									
	I_{IL2}		4	0.4 V										2.0 V									
			5		0.4 V																		
			6			0.4 V																	
			7				0.4 V																
			8						0.4 V						0.4 V								
			9						0.8 V						0.8 V								
			10								0.4 V					0.4 V							
			11								0.8 V					0.8 V							
			12																		0.4 V	0.4 V	
			13																		0.8 V	0.8 V	
			14																				0.4 V
			15																				4.9 V
			16																				
			17																				
			18																				
			19																				
			20																				
			21																				
			22																				
			23																				
			24																				
			25																				
			26																				
			27																				
			28																				
			29																				
			30																				
	I_{IH2}		31	2.4 V																			
			32		2.4 V																		
			33			2.4 V																	
			34				2.4 V																
			35						2.4 V						2.4 V								
			36						0.8 V						2.0 V								
			37								2.4 V					2.4 V							
			38								0.8 V					2.0 V							
			39														2.4 V				2.4 V	2.4 V	
			40														2.0 V				2.0 V	2.0 V	
			41																				2.4 V
			42																				4.9 V
			43																				
			44																				
			45																				
			46																				
			47																				
			48																				
			49																				
			50																				
			51																				
			52																				
			53																				
			54																				

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TABLE III. Group A inspection for device types 01 and 02 - Continued.
Terminal conditions (pins not designated are open).

Subgroup	Symbol	MIL-STD-883 method	Test no.	1 D1	2 D3	3 D5	4 D7	5 $\overline{INT}$	6 IEI	7 IEO	8 $\overline{M1}$	9 V_{CC}	10 W/$\overline{RDYA}$	11 SYNCA	12 RxDA	13 $\overline{RxCA}$	14 $\overline{TxCA}$	15 TxDA	16 $\overline{DTRA}$	17 $\overline{RTSA}$	18 $\overline{CTSA}$	19 $\overline{DCDA}$	20 CLK
1 T_C = +25°C	I_{IH2}		55						0.8 V		0.8 V	5.5 V		2.0 V	2.0 V	2.0 V					2.0 V	2.0 V	4.9 V
	I_{IH2}		56																				
	I_{IH2}		57																				
	I_{LOL}		58					0.4 V									2.0 V						
	I_{LOL}		59										0.4 V										
	I_{LOL}		60																				
	I_{LOH}		61					2.4 V															
	I_{LOH}		62										2.4 V										
	I_{LOH}		63																				
2 T_C = +125°C	Repeat tests 1 through 63 at +125°C with same test limits and conditions except the limit for test 1 shall be 180 mA.																						
3 T_C = -55°C	Repeat tests 1 through 63 at -55°C with same test limits and conditions except the limit for test 1 shall be 91 mA.																						
4 T_C = +25°C	C_{I1}		64	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	1/ GND
	C_{I2}		65						1/ GND		GND 1/ GND	GND GND 1/ GND		1/ GND	1/ GND	1/ GND	1/ GND				1/ GND	1/ GND		
			66																					
			67																					
			68																					
			69																					
			70																					
			71																					
			72																					
			73																					
			74																					
			75																					
			76																					
			77																					
			78																					
			79																					
			80																					
			81																					
			82																					
			83																					
			84																					
	C_O		85	1/ GND	1/ GND	1/ GND	1/ GND	1/ GND		1/ GND			1/ GND					1/ GND GND GND	1/ GND GND	1/ GND			
			86																					
			87																					
			88																					
			89																					
			90																					
			91																					
			92																					
			93																					
			94																					
			95																					

See footnote at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.
Electrical conditions (pins not designated are open).

Test no.	21 RESET	22 $\overline{DCDB}$	23 $\overline{CTSB}$	24 $\overline{RTSB}$	25 $\overline{DTRB}$	26 TxDB	27 $\overline{TxCB}$	28 $\overline{RxCB}$	29 RxDB	30 W/$\overline{RDYB}$	31 GND	32 $\overline{RD}$	3 C/$\overline{D}$	34 B/$\overline{A}$	35 $\overline{CE}$	36 $\overline{IORQ}$	37 D6	38 D4	39 D2	40 D0	Measured terminal	Test Limits Min	Test Limits Max	Unit
55	0.8 V	2.0 V	2.0 V				2.0 V	2.0 V	2.0 V		GND	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V		2.4 V			38	-10	10	µA
56																			2.4 V		39			
57																				2.4 V	40			
58																					5			
59																					10			
60										0.4 V											30			
61																					5			
62																					10			
63										2.4 V											30			
64	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	20		40	pF
65																					6		5	
66																					8			
67																					9			
68																					11			
69																					12			
70																					13			
71																					14			
72																					18			
73	1/ GND																				19			
74		1/ GND																			21			
75			1/ GND																		22			
76							1/ GND														23			
77								1/ GND													27			
78									1/ GND												28			
79												1/ GND									29			
80													1/ GND								32			
81														1/ GND							33			
82															1/ GND						34			
83																1/					35			
84																					36			
85																GND					1		10	
86																					2			
87																					3			
88																					4			
89																					5			
90																					7			
91																					10			
92																					15			
93																					16			
94				1/																	17			
95																					24			

See footnote at end of table.

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TABLE III. Group A inspection for device types 01 and 02 - Continued.
Terminal conditions (pins not designated are open).

Subgroup	Symbol	MIL-STD-883 method	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
				D1	D3	D5	D7	$\overline{INT}$	IEI	IEO	$\overline{WI}$	V_{CC}	$\overline{W/RDYA}$	$\overline{SYNCA}$	RxDA	$\overline{RxCA}$	$\overline{TxCA}$	TxDA	$\overline{DTRA}$	$\overline{RTSA}$	$\overline{CTSA}$	$\overline{DCDA}$	CLK
4 T_C = +25°C	C_O		96 97 98 99 100 101 102	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND
7 T_C = +25°C	3/, 4/, 6/	3011	See 6.1 (table V)																				
8	3/, 4/, 6/	3014	Same tests, terminal conditions and limits as for subgroup 7 except T_C = +125°C and T_C = -55°C.																				
9 T_C = +25°C	3/, 4/, 5/	3014																					
10 T_C = +125°C	3/, 4/, 5/	3014	Repeat subgroup 9 at +125°C.																				
11 T_C = -55°C	3/, 4/, 5/	3014	Repeat subgroup 9 at -55°C.																				

See footnotes at end of table.

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Test no.	1 RESET	22 DCDB	23 CTSB	24 RTSB	25 DTRB	26 TxDB	27 TxCB	28 RxCB	29 RxDB	30 W/RDYB	31 GND	32 RD	33 C/D	34 B/A	35 CE	36 IORQ	37 D6	38 D4	39 D2	40 D0	Measured terminal	Test Limits Min	Test Limits Max	Unit
96	GND	GND	GND	GND	1/ GND	GND 1/ GND	GND	GND	GND	GND GND 1/ GND	GND	GND	GND	GND	GND	GND	GND 1/ GND GND GND	GND 1/ GND GND	GND 1/ GND	GND 1/	25 26 30 37 38 39 40		10	pF
97																								
98																								
99																								
100																								
101																								
102																								

1/ All pins not designated are open.

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TABLE III. Group A inspection for device types 01 and 02.
Terminal conditions (pins not designated are open).

Subgroup 7/, 8/	Symbol	Test no.	Terminal(s)	9/ Figure number	10/ Reference number	Limits			
						Device types			
						01		02	
						Minimum	Maximum	Minimum	Maximum
12 T_C = +25°C and +125°C	t_{PHL1} t_{PLH1}	103	DO-D7 DO-D7	6	8		220		240
	t_{PHZ1} t_{PLZ1}	104	DO-D7 DO-D7		10		110		230
	t_{PHL2} t_{PLH2}	105	DO-D7 DO-D7		11		160		340
	t_{PHL3} t_{PLH3}	106	TxD TxD		27		300		400
	t_{PHL4}	107	IEO		14	190		300	
	t_{PLH4}	108	IEO		15		100		150
	t_{PLH5}	109	IEO		16		100		150
	t_{PHL6}	110	$\overline{INT}$		17		200		200
	t_{PHL7}	111	$\overline{W/RDY}$		18		210		300
	t_{PHL8}	112	$\overline{W/RDY}$		19		120		120
	t_{PLZ2}	113	$\overline{W/RDY}$		20		130		150
	t_{PHL9}	114	$\overline{W/RDY}$		28	5 2/	9 2/	5 2/	9 2/
	t_{PHL10}	115	$\overline{INT}$		29	5 2/	9 2/	5 2/	9 2/
	t_{PHL11}	116	$\overline{W/RDY}$		35	10 2/	13 2/	10 2/	13 2/
	t_{PHL12}	117	$\overline{INT}$		36	10 2/	13 2/	10 2/	13 2/
	t_{PHL13}	118	$\overline{SYNC}$		37	4 2/	7 2/	4 2/	7 2/

1/ The capacitance measurement shall be made between the indicated terminal and ground at a frequency of 1 MHz. The dc bias of the measurement instrument shall be less than ±0.1 V. The ac amplitude shall be less than 50 mV rms.

2/ Clock cycles (system clock).

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For functional tests (and subgroups 9, 10, and 11), the input/output conditions A, B, C, and D appear in table IV.

4/ Table VI (see 6.1) functional tests shall be repeated for input/output conditions A through D of table IV.

5/ All subgroups 9, 10, and 11 are run concurrently with subgroups 7 and 8.

6/ V_{OH} and V_{OL} are measured as part of subgroups 7 and 8.

7/ Where propagation delay tests for subgroup 12 indicate a group of pins, and where two or more propagation delays are specified for a test, only the reading for the pin with the worst delay requires recording. In some cases this will be a minimum reading, while for others it will be a maximum reading, as indicated in the table.

8/ Table VI functional tests shall be repeated for input/output conditions A through D of table IV for tests 103 through 118.

9/ Figure 6 represents the electrical performance characteristics associated with the ac limits shown in table III.

10/ The reference number refers to the position where the parameter being tested appears on figure 6.

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TABLE IV. Input/output conditions for table III subgroups 7 through 11.
Device type 01

Symbol	Reference 2/	Terminal(s)	A	B	C	D	Unit
V_{CC}		V_{CC}	5.5	4.5	4.5	4.5	V
V_{IH2}		Logic inputs	2.0	2.0	2.0	2.0	V
V_{IL2}		Logic inputs	0.8	0.8	0.8	0.8	V
V_{IH1}		CLK	4.9	3.9	3.9	3.9	V
V_{IL1}		CLK	0.45	0.45	0.45	0.45	V
V_{OH}		Outputs	2.4	2.4	2.4	2.4	V
V_{OL}		Outputs	0.4	0.4	0.4	0.4	V
t_{CYC1}	1	CLK	250	250	2000	2000	ns
t_{PWL1}	5	CLK	105	105	105		ns
t_{PWH1}	2	CLK				105	ns
t_{CYC2}	24	$\overline{TxC}$	400	400	8000	8000	ns
t_{PWL3}	25	$\overline{TxC}$	180	180	180		ns
t_{PWH3}	26	$\overline{TxC}$				180	ns
t_{CYC3}	30	$\overline{RxC}$	400	400	8000	8000	ns
t_{PWL4}	31	$\overline{RxC}$	180	180	180		ns
t_{PWH4}	32	$\overline{RxC}$				180	ns
t_{SHL2} t_{SLH2}	7	Control inputs	115	115	115	115	ns
t_{SHL3} t_{SLH3}	9	D0-D7	50	50	50	50	ns
t_{SHL7}	38	$\overline{SYNC}$	-100	-100	-100	-100	ns
t_{PHL1} t_{PLH1}	8	D0-D7	220	220	220	220	ns
t_{PHL6}	17	$\overline{INT}$	200	200	200	200	ns
t_{PHL8}	19	$\overline{W/RDY}$	120	120	120	120	ns
t_{PHL3} t_{PLH3}	27	TxD	300	300	300	300	ns
t_{PHL9}	28	$\overline{W/RDY}$	5 9	5 9	5 9	5 9	1/ I/
t_{PHL10}	29	$\overline{INT}$	5 9	5 9	5 9	5 9	1/ I/
t_{PHL11}	35	$\overline{W/RDY}$	10 13	10 13	10 13	10 13	1/ I/

See footnotes at end of table.

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TABLE IV. Input/output conditions for table III subgroups 7 through 11 - Continued.
Device type 01

Symbol	Reference 2/	Terminal(s)	A	B	C	D	Unit
t_{PHL12}	36	$\overline{INT}$	10 13	10 13	10 13	10 13	1/ I/
t_{PHL13}	37	$\overline{SYNC}$	4 7	4 7	4 7	4 7	1/ I/
t_{PHZ1} t_{PLZ1}	10	D0-D7	110	110	110	110	ns
t_{PHL2} t_{PLH2}	11	D0-D7	160	160	160	160	ns
t_{PHL4}	14	IEO	190	190	190	190	ns
t_{PLH4}	15	IEO	100	100	100	100	ns
t_{PLH5}	16	IEO	100	100	100	100	ns
t_{PHL7}	18	$\overline{W/RDY}$	210	210	210	210	ns
t_{PZL2}	20	$\overline{W/RDY}$	130	130	130	130	ns
t_{SHL1} t_{SLH1}	6	$\overline{CE}$, C/$\overline{D}$, D/$\overline{A}$	145	145	145	145	ns
t_{HL4} $_{H4}$	33	$\overline{RxC}$	0	0	0	0	ns
$_{HL5}$ t_{SLH5}	12	$\overline{M1}$	90	90	90	90	ns
t_{SHL6}	13	$\overline{IORQ}$	140	140	140	140	ns
t_{HHL1} t_{HLH1}	34	RxD	140	140	140	140	ns
t_{HHL2} t_{HLH2}	21	All unspecified hold times	0	0	0	0	ns

See footnotes at end of table.

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TABLE IV. Input/output conditions for table III.
Device type 02

Symbol	Reference 2/	Terminal(s)	A	B	C	D	Unit
V_{CC}		V_{CC}	5.5	4.5	4.5	4.5	V
V_{IH2}		Logic inputs	2.0	2.0	2.0	2.0	V
V_{IL2}		Logic inputs	0.8	0.8	0.8	0.8	V
V_{IH1}		CLK	4.9	3.9	3.9	3.9	V
V_{IL1}		CLK	0.45	0.45	0.45	0.45	V
V_{OH}		Outputs	2.4	2.4	2.4	2.4	V
V_{OL}		Outputs	0.4	0.4	0.4	0.4	V
t_{CYC1}	1	CLK	400	400	2000	2000	ns
t_{PWL1}	5	CLK	170	170	170		ns
t_{PWH1}	2	CLK				170	ns
t_{CYC2}	24	$\overline{TxC}$	400	400	8000	8000	ns
t_{PWL3}	25	$\overline{TxC}$	180	180	180		ns
t_{PWH3}	26	$\overline{TxC}$				180	ns
t_{CYC3}	30	$\overline{RxC}$	400	400	8000	8000	ns
t_{PWL4}	31	$\overline{RxC}$	180	180	180		ns
t_{PWH4}	32	$\overline{RxC}$				180	ns
t_{SHL2} t_{SLH2}	7	Control inputs	240	240	240	240	ns
t_{SHL3} t_{SLH3}	9	D0-D7	50	50	50	50	ns
t_{SHL7}	38	$\overline{SYNC}$	-100	-100	-100	-100	ns
t_{PHL1} t_{PLH1}	8	D0-D7	240	240	240	240	ns
t_{PHL6}	17	$\overline{INT}$	200	200	200	200	ns
t_{PHL8}	19	$\overline{W/RDY}$	120	120	120	120	ns
t_{PHL3} t_{PLH3}	27	TxD	300	300	300	300	ns
t_{PHL9}	28	$\overline{W/RDY}$	5 9	5 9	5 9	5 9	1/ 1/
t_{PHL10}	29	$\overline{INT}$	5 9	5 9	5 9	5 9	1/ 1/
t_{PHL11}	35	$\overline{W/RDY}$	10 13	10 13	10 13	10 13	1/ 1/

See footnotes at end of table.

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TABLE IV. Input/output conditions for table III subgroups 7 through 11 - Continued.
Device type 02

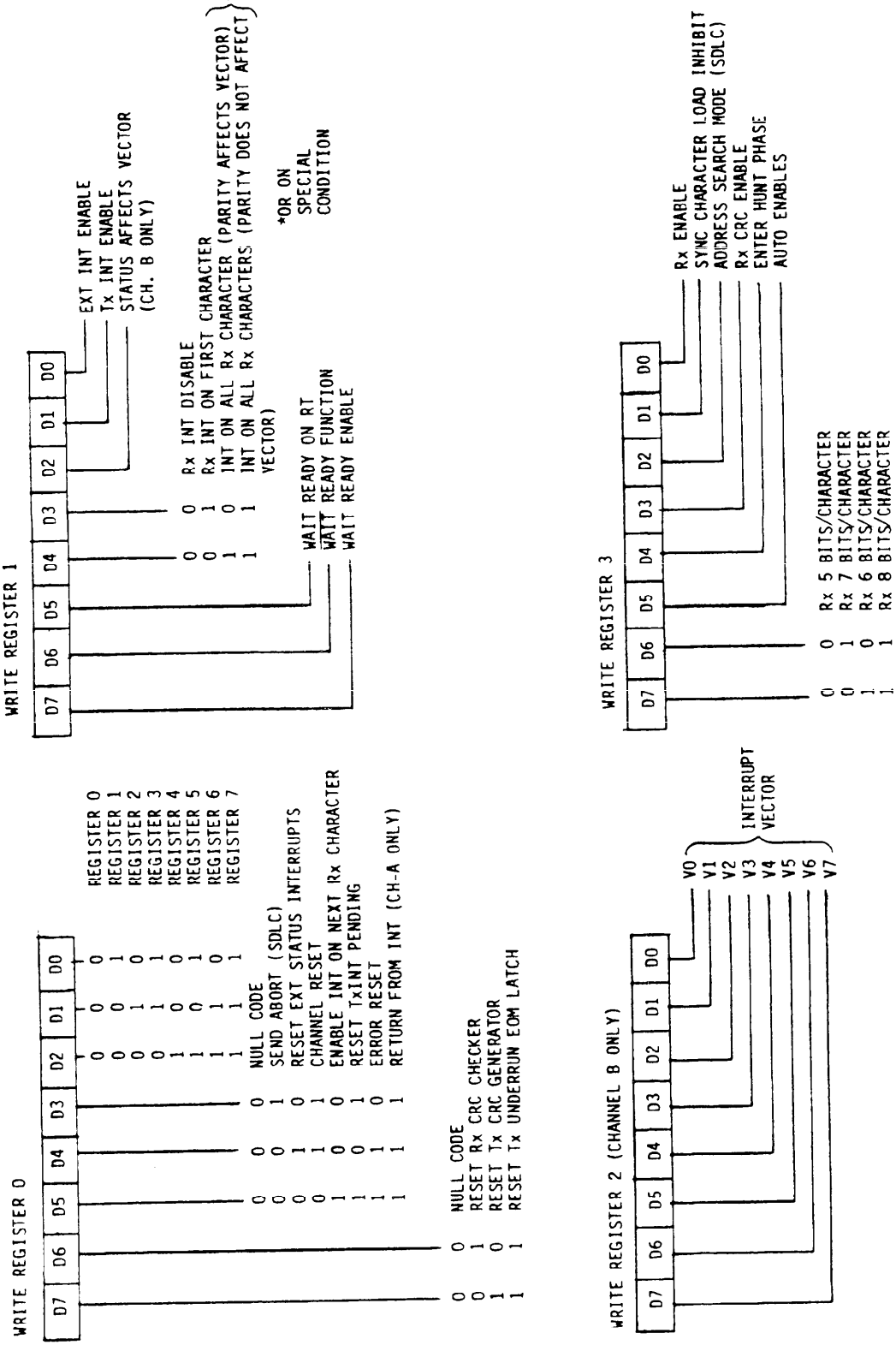
Symbol	Reference 2/	Terminal(s)	A	B	C	D	Unit
t_{PHL12}	36	$\overline{INT}$	10 13	10 13	10 13	10 13	1/ 1/
t_{PHL13}	37	$\overline{SYNC}$	4 7	4 7	4 7	4 7	1/ 1/
t_{PHZ1} t_{PLZ1}	10	D0-D7	230	230	230	230	ns
t_{PHL2} t_{PLH2}	11	D0-D7	340	340	340	340	ns
t_{PHL4}	14	IEO	300	300	300	300	ns
t_{PLH4}	15	IEO	150	150	150	150	ns
t_{PLH5}	16	IEO	150	150	150	150	ns
t_{PHL7}	18	$\overline{W/RDY}$	300	300	300	300	ns
t_{PLZ2}	20	$\overline{W/RDY}$	150	150	150	150	ns
t_{SHL1} t_{SLH1}	6	$\overline{CE}$, C/$\overline{D}$, D/$\overline{A}$	160	160	160	160	ns
$_{HL4}$ t_{SLH4}	33	$\overline{RxC}$	0	0	0	0	ns
t_{SHL5} t_{SLH5}	12	$\overline{M1}$	210	210	210	210	ns
t_{SHL6}	13	$\overline{IORQ}$	200	200	200	200	ns
t_{HHL1} t_{HLH1}	34	RxD	140	140	140	140	ns
t_{HHL2} t_{HLH2}	21	All unspecified hold times	0	0	0	0	ns

1/ Minimum/maximum multiples of clock cycles. The indicated signals must change state within the specified window.

2/ The reference numbers refer to the position where the parameter being tested appears on figure 6.

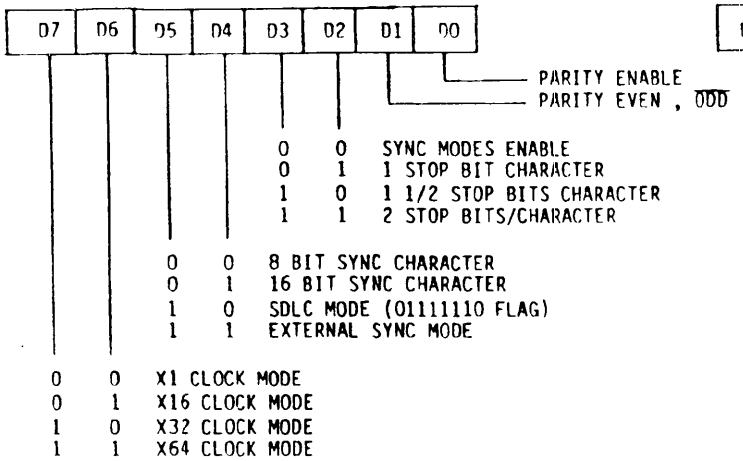
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TABLE V. Register definitions.

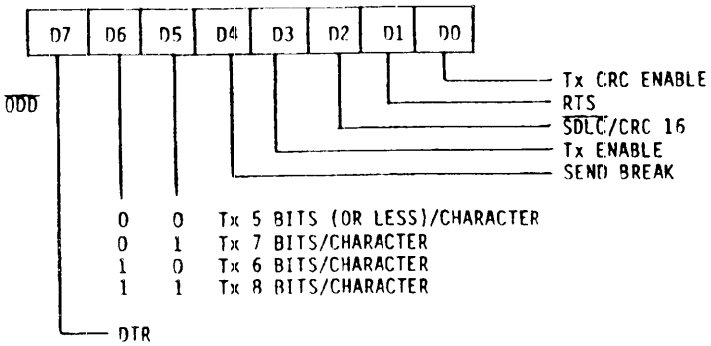


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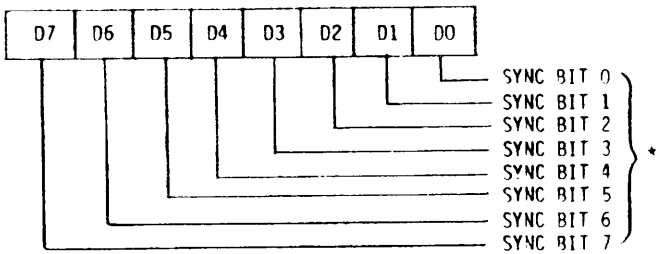
WRITE REGISTER 4



WRITE REGISTER 5

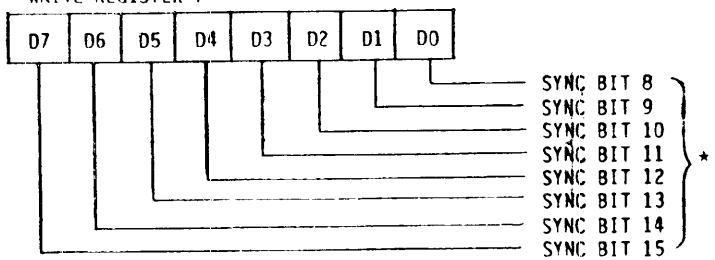


WRITE REGISTER 6



*ALSO SDLC ADDRESS FIELD

WRITE REGISTER 7

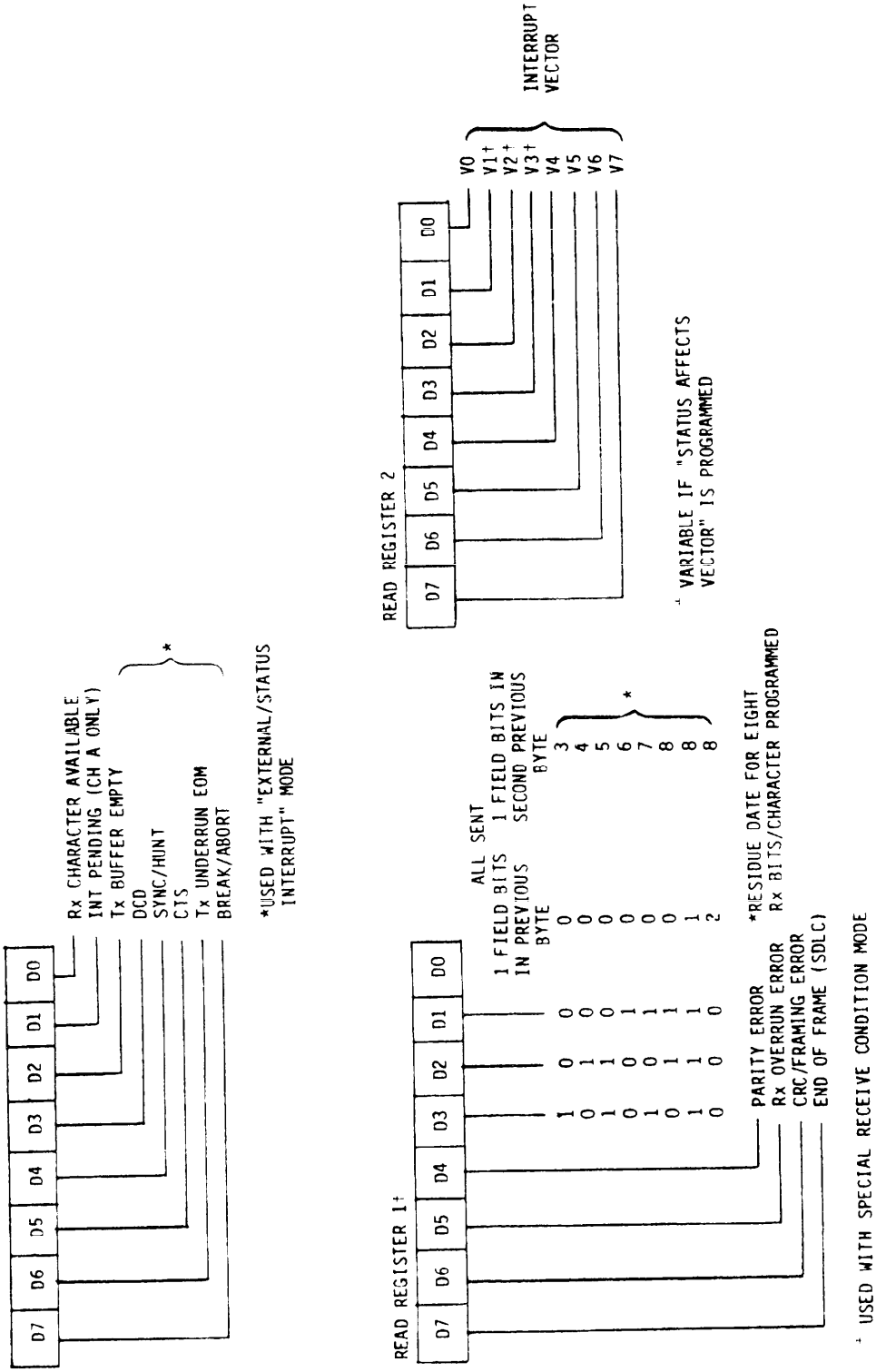


* FOR SDLC IT MUST BE PROGRAMMED TO "01111110" FOR FLAG RECOGNITION

MIL-M-38510/501A(USAF)

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TABLE V. Register definitions - Continued.



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4.. Methods of inspection. Methods of inspection shall be as specified in the appropriate table and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 High-voltage (V_{ZAP}) test of input protection circuits. One input terminal of the device under test (DUT) shall be subjected to a voltage pulse of 400 V from a 100 pF source in the following test sequence.

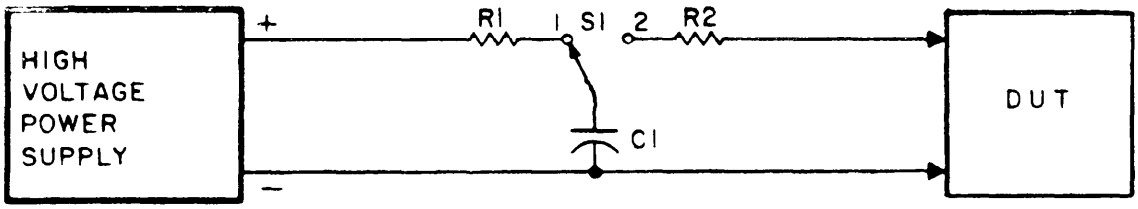
a. Measure I_{IH} and I_{IL} at one input terminal of the device under test at 25°C. These measurements shall be made in accordance with table III herein. The test limits for a single terminal measurement of I_{IH} and I_{IL} shall be as specified in table III.

b. Apply the test voltage (V_{ZAP} = 400 V) to the same terminal selected for leakage current measurements. Apply V_{ZAP} in a 2-pulse sequence as follows and in accordance with the test circuit of figure 7.

(1) Input (-) to GND.

(2) Input (+) to GND.

c. Within 24 hours repeat the I_{IH} and I_{IL} measurements on the same terminal as performed above. At this time a DUT exhibiting leakage currents in excess of the specified limits is defective.



V_{ZAP} = 400 V measured at the device input terminal.

10 MΩ ≤R1 ≤50 MΩ.

R2 = 1.5 kΩ.

C1 = 100 pF.

S1 = Hg-wetted "bounceless" relay break before make type. All pins of DUT not associated with V_{ZAP} test shall be open (NC).

FIGURE 7. High voltage (V_{ZAP}) test circuits.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Test vector table. Table VI, which forms a part of this specification, is not printed herein because of its extreme length and complexity; and, even when presented as hard copy, it requires conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be obtained in a form compatible with test system architecture (magnetic tape format usually preferred). The preparing activity (or DESC, as its agent) may be consulted for information on obtaining such output. For reference purposes, Rome Air Development Center (RADC-RBE-2) and Defense Electronics Supply Center (DESC-ECS) each maintain a limited quantity of dated, hard copy printouts of this table.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the data (see 4.4) pertinent to the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirement for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the government.

6.4 Terms and definitions. The abbreviations, terms, symbols, and definitions used herein (including terms and symbols for device terminals) are defined in MIL-M-38510, MIL-STD-1331, and as follows:

$B/\overline{A}$ (Channel A or B select)	Input, high selects Channel B. This input defines which channel is accessed during a data transfer between the CPU and the SIO. Address bit A_0 from the CPU is often used for the selection function.
$C/\overline{D}$ (Control or Data Select)	Input, high selects Control. This input defines the the type of information transfer performed between the CPU and the SIO. A high at this input during a CPU write to the SIO causes the information on the data bus to be interpreted as a command for the channel selected by the $B/\overline{A}$. A low at $C/\overline{D}$ means that the information on the data bus is data. Address bit A_1 is often used for this function.
$\overline{CE}$ (Chip enable)	Input, active low. A low level at this input enables the SIO to accept command or data input from the CPU during a write cycle, or to transmit data to the CPU during a read cycle.

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CLK (System Clock)	Input. The SIO uses the standard System Clock to synchronize internal signals. This is a single-phase clock.
$\overline{CTSA}$, $\overline{CTSB}$ (Clear to Send)	Inputs, active low. When programmed as Auto Enables, a low on these inputs enables the respective transmitter. If not programmed as Auto Enables, these inputs may be programmed as general-purpose inputs. Both inputs are Schmitt-trigger buffered to accommodate slow-rise time signals. The SIO detects pulses on these inputs and interrupts the CPU on both logic level transitions. The Schmitt-trigger buffering does not guarantee a specified noise-level margin.
$D_0 - D_7$ (System Data Bus)	Bidirectional, 3-state. The system data bus transfers data and commands between the CPU and the SIO. D_0 is the least significant bit.
$\overline{DCDA}$, $\overline{DCDB}$ (Data Carrier Detect)	Inputs, active low. These pins function as receiver enables if the SIO is programmed for Auto Enables; otherwise they may be used as general-purpose input pins. Both pins are Schmitt-trigger buffered to accommodate slow-rise time signals. The SIO detects pulses on these pins and interrupts the CPU on both logic level transitions. Schmitt-trigger buffering does not guarantee a specific noise-level margin.
$\overline{DTRA}$, $\overline{DTRB}$ (Data Terminal Ready)	Outputs, active low. These outputs follow the state programmed into SIO. They can also be programmed as general-purpose outputs.
IEI (Interrupt Enable In)	Input, active high. This signal is used with IEO to form a priority daisy chain when there is more than one interrupt-driven device. A high on this line indicates that no other device of high priority is being serviced by a CPU interrupt service routine.
IEO (Interrupt Enable Out)	Output, active high. IEO is high only if IEI is high and the CPU is not servicing an interrupt from this SIO. Thus, this signal blocks lower priority devices from interrupting while a higher priority device is being serviced by its CPU interrupt service routine.
$\overline{INT}$ (Interrupt Request)	Output, open drain, active low. When the SIO is requesting an interrupt, it pulls $\overline{INT}$ low.
$\overline{IORQ}$ (Input/Output Request)	Input from CPU, active low. $\overline{IORQ}$ is used in conjunction with B/$\overline{A}$, C/$\overline{D}$, $\overline{CE}$ and $\overline{RD}$ to transfer commands and data between the CPU and the SIO. When $\overline{CE}$, $\overline{RD}$ and $\overline{IORQ}$ are all active, the channel selected by B/$\overline{A}$ transfers data to the CPU (a read operation). When $\overline{CE}$ and $\overline{IORQ}$ are active, but $\overline{RD}$ is inactive, the channel selected by B/$\overline{A}$ is written to by the CPU with either data or control information as specified by C/$\overline{D}$. As mentioned previously, if $\overline{IORQ}$ and $\overline{M1}$ are active simultaneously, the CPU is acknowledging an interrupt and the SIO automatically places its interrupt vector on the CPU data bus if it is the highest priority device requesting an interrupt.

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$\overline{M1}$ (Maching Cycle)	Input from CPU, active low. When $\overline{M1}$ is active and $\overline{RD}$ is also active, the CPU is fetching an instruction from memory; when $\overline{M1}$ is active while $\overline{IORQ}$ is active, the SIO accepts $\overline{M1}$ and $\overline{IORQ}$ as an interrupt acknowledge if the SIO is the highest priority device that has interrupted the CPU.
$\overline{RxCA}$, $\overline{RxCB}$ (Receiver Clocks)	Inputs. Receive data is sampled on the rising edge of $\overline{RxC}$. The Receive Clocks may be 1, 16, 32 or 64 times the data rate in asynchronous modes. These clocks may be driven by the CTC Counter Timer Circuit for programmable baud rate generation. Both inputs are Schmitt-trigger buffered (no noise level margin is specified).
$\overline{RD}$ (Read Cycle Status)	Input from CPU, active low. If $\overline{RD}$ is active, a memory or I/O read operation is in progress. $\overline{RD}$ is used with B/$\overline{A}$, $\overline{CE}$ and $\overline{IORQ}$ to transfer data from the SIO to the CPU.
RxDA, RxDB (Receive Data)	Inputs, active high. Serial data at TTL levels.
$\overline{RESET}$ (Reset)	Input, active low. A low $\overline{RESET}$ disables both receivers and transmitters, forces TxDA and TxDB marking, forces the modem controls high and disables all interrupts. The control registers must be rewritten after the SIO is reset and before data is transmitted or received.
$\overline{RTSA}$, $\overline{RTSB}$ (Request to Send)	Outputs, active low. When the RTS bit in Write Register 5 is set, the $\overline{RTS}$ output goes low. When the RTS bit is reset in the Asynchronous mode, the output goes high after the transmitter is empty. In Synchronous modes, the $\overline{RTS}$ pin strictly follows the state of the RTS bit. Both pins can be used as general-purpose outputs.
$\overline{SYNCA}$ (Synchronization)	Input/output, active low. This pin can act either as input or output. In the asynchronous receive mode, it is an input similar to $\overline{CTS}$ and $\overline{DCD}$. In this mode, the transitions on this line affect the state of the Sync/Hunt status bits in Read Register 0, but have no other function. In the External Sync mode, this line also acts as input. When external synchronization is achieved, $\overline{SYNC}$ must be driven low on the second rising edge of $\overline{RxC}$ after that rising edge of $\overline{RxC}$ on which the last bit of the sync character was received. In other words, after the sync pattern is detected, the external logic must wait for two full Receive Clock cycles to activate the $\overline{SYNC}$ input. Once $\overline{SYNC}$ is forced low, it should be kept low until the CPU informs the external synchronization detect logic that synchronization has been lost or a new message is about to start. Character assembly begins on the rising edge of $\overline{RxC}$ that immediately precedes the falling edge of $\overline{SYNC}$ in the External Sync mode. In the internal synchronization mode (Monosync and Bisync), this pin acts as an output that is active during the part of the receive clock ($\overline{RxC}$) cycle in which sync characters are recognized. The sync condition is not latched, so this output is active each time a sync pattern is recognized, regardless of character boundaries.

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TxCA, $\overline{\text{TxCB}}$ (Transmitter Clocks)	TxD changes from the falling edge of $\overline{\text{TxC}}$. In asynchronous modes, the Transmitter Clocks may be 1, 16, 32 or 64 times the data rate; however, the clock multiplier for the transmitter and the receiver must be the same. The Transmit Clock inputs are Schmitt-trigger buffered for relaxed rise- and fall-time requirements (no noise level margin is specified). Transmitter Clocks may be driven by the CTC Counter Timer Circuit for programmable baud rate generation.
TxDA, TxDB (Transmit Data)	Outputs, active high. Serial data at TTL levels.
W/RDYA, $\overline{\text{W/RDYB}}$ (Wait/Ready A, Wait/Ready B)	Outputs, open drain, when programmed for Wait function; driven high and low when programmed for Ready function. These dual-purpose outputs may be programmed as Ready lines for a DMA controller or as Wait lines that synchronize the CPU to the SIO data rate. The reset state is open drain.

6.5 Logistic support. Lead material and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish "C" (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Handling. MOS microprocessors peripherals should be handled with certain precautions to avoid damage due to the discharge of accumulated static charge. These NMOS devices are fabricated with a silicon gate technology, including input protection, which reduces the susceptibility to damage, however, the following handling practices are recommended.

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Handling of devices by the leads should be avoided.

d. Devices should be stored in conductive carriers.

e. Avoid uses of plastic, rubber, or silk in MOS areas.

6.7 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.3) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to maintain the generic data for a period of not less than 36 months from the date of shipment.

6.8 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military-device type	Generic-industry type
01	Z-80A SIO/2
02	Z-80 SIO/2

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6.9 Ordering guidance. Since the qualification requirements have been removed from the specification, orders may be placed immediately.

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 17

User activities:
Air Force - 11, 19, 85, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/481A(USAF)	Microcircuits, Digital, N-Channel, Silicon Gate Monolithic Dual Serial Input/Output Controller

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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